PLC Counter Instructions

Pages 2–4 (Lecture Notes)

1. Electronic Counters

Electronic counters are used in PLC systems to count the occurrence of events such as pushbutton presses, sensor activations, or machine operations.

Types of electronic counters:

- **Up-counter**: Counts upward (0, 1, 2, 3, ...).
- **Down-counter**: Counts downward from a preset value.
- **Up/Down counter**: Can count both upward and downward.

2. Counter Representation in PLC Ladder Logic

There are two common methods to represent counters in PLC programs:

- **Coil-formatted counters**
- **Block-formatted counters**

3. Coil-Formatted Counter Instruction

In coil format, the counter appears as a ladder logic instruction similar to an output coil.

Operation

- The counter increments when the rung transitions from **FALSE to TRUE**.
- Each valid transition increases the accumulated value by **1**.
- Continuous ON state does **not** cause repeated counting.

Counter Completion Condition

When the **accumulated count equals the preset count**, the counter output becomes energized (logic 1).

Parts of a Counter Instruction

- Counter type (Up or Down)
- Counter address

- Preset value
- Accumulated value

4. Counter Reset Instruction

In coil-formatted counters, a separate reset instruction is used to clear the accumulated count.

Reset rule:

- The reset instruction must reference the **same counter address** as the counter itself.

After reset:

- Accumulated value is set to a predetermined value (usually zero).
- Counting resumes after the reset condition becomes false.

5. Block-Formatted Counter Instruction

Block-formatted counters represent counters as function blocks with explicit inputs and stored values.

Instruction block contains:

- Counter type (Up or Down)
- Preset value
- Accumulated (current) value

Input conditions:

- Count input
- Reset input

6. Edge Triggering in PLC Counters

PLC counters operate only on the **leading edge** of the input signal.

Valid Counting Condition

- Input transition from **OFF to ON** triggers counting.

Invalid Counting Conditions

- ON to OFF transition (trailing edge)
- Continuous ON signal

This behavior prevents false multiple counts and ensures accurate event counting.

7. Counting Sequence of Counters

Up-Counter

- Accumulated value starts from zero.
- Increments by one for each valid input pulse.
- Stops counting when accumulated value equals preset value.

Down-Counter

- Accumulated value starts from preset value.
- Decrements by one for each valid input pulse.
- Output is energized when the target value is reached.

Counter Reset Effect

- Clears accumulated value.
- Allows the counter to restart the counting sequence.